

**STUDENT DEPRESSION ANALYSIS: UNDERSTANDING THE HIDDEN
PATTERNS BEHIND ACADEMIC PRESSURE AND MENTAL HEALTH**

BY

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INTRODUCTION

Mental health has become a growing concern among students, especially in competitive academic environments. For the purpose of this analysis, I assumed the dataset represented students from Imperial college London, exploring how academic performance, lifestyle, and personal circumstances shape mental health outcomes.

The Student Depression Dataset, sourced from Kaggle, offers a detailed view of these variables, including demographic information (age, gender), academic performance (CGPA, study satisfaction), and lifestyle factors (sleep duration, diet, work pressure).

a. Objective:

To identify the key factors that increase students' risk of depression and guide strategies that can help improve overall student well-being and academic performance.

Stakeholders:

- University Management and Program Directors
- Student Affairs and Counseling Teams
- Educational Researchers and Data Scientists

b. Problem Being Addressed:

What are the primary lifestyle, academic, and social factors contributing to depression among students—and how can universities address them?

c. Key Datasets and Methodologies:

The analysis was carried out using Microsoft Excel for cleaning, transformation, and exploratory analysis. Visualizations and insights were derived through structured comparison across gender, age groups, degree programs, and cities.

THE STORY OF DATA

a. Data Source: Kaggle—*Student Depression Dataset*

Year Covered: 2022

Total Records: 27,902

Total Columns: 18

Each record represents an individual student and includes attributes such as:

- Age and Gender
- City and Profession
- CGPA (Cumulative Grade Point Average)
- Sleep Duration
- Work Pressure and Academic Pressure
- Study and Job Satisfaction
- Dietary Habits
- Depression Status (target variable: Yes / No)

id	Gender	Age	City	Profession	Academic Pressu	Work Pressure	CGPA	Study Satisfac	Job Satisfaction	Sleep Duration	Dietary Habits	Degree	Have you ever h	Work/Study	Financial Stres	Family His	Depression
2	Male	33	Visakhapat	Student	5	0	8.97	2	0	5-6 hours	Healthy	B.Pharm	Yes	3	1	No	1
8	Female	24	Bangalore	Student	2	0	5.9	5	0	5-6 hours	Moderate	BSc	No	3	2	Yes	0
26	Male	31	Srinagar	Student	3	0	7.03	5	0	Less than 5 hour	Healthy	BA	No	9	1	Yes	0
30	Female	28	Varanasi	Student	3	0	5.59	2	0	7-8 hours	Moderate	BCA	Yes	4	5	Yes	1
32	Female	25	Jaipur	Student	4	0	8.13	3	0	5-6 hours	Moderate	M.Tech	Yes	1	1	No	0
33	Male	29	Pune	Student	2	0	5.7	3	0	Less than 5 hour	Healthy	PhD	No	4	1	No	0
52	Male	30	Thane	Student	3	0	9.54	4	0	7-8 hours	Healthy	BSc	No	1	2	No	0
56	Female	30	Chennai	Student	2	0	8.04	4	0	Less than 5 hour	Unhealthy	Class 12	No	0	1	Yes	0
59	Male	28	Nagpur	Student	3	0	9.79	1	0	7-8 hours	Moderate	B.Ed	Yes	12	3	No	1
62	Male	31	Nashik	Student	2	0	8.38	3	0	Less than 5 hour	Moderate	LLB	Yes	2	5	No	1
83	Male	24	Nagpur	Student	3	0	6.1	3	0	5-6 hours	Moderate	Class 12	Yes	11	1	Yes	1
91	Male	33	Vadodara	Student	3	0	7.03	4	0	Less than 5 hour	Healthy	BE	Yes	10	2	Yes	0
94	Male	27	Kalyan	Student	5	0	7.04	1	0	Less than 5 hour	Moderate	M.Tech	No	10	1	Yes	1
100	Female	19	Rajkot	Student	2	0	8.52	4	0	Less than 5 hour	Unhealthy	Class 12	No	6	2	Yes	0
103	Female	19	Kalyan	Student	5	0	5.64	5	0	Less than 5 hour	Moderate	Class 12	Yes	4	5	Yes	1
106	Male	29	Srinagar	Student	3	0	8.58	3	0	More than 8 hour	Moderate	M.Tech	Yes	10	2	Yes	1
120	Male	25	Nashik	Student	5	0	6.51	2	0	Less than 5 hour	Unhealthy	M.Ed	Yes	2	5	Yes	1
132	Female	20	Ahmedaba	Student	5	0	7.25	3	0	5-6 hours	Healthy	Class 12	Yes	10	3	No	1
139	Male	19	Chennai	Student	2	0	7.83	2	0	7-8 hours	Unhealthy	Class 12	No	6	3	No	0
145	Male	25	Kalyan	Student	3	0	9.93	3	0	5-6 hours	Moderate	B.Ed	No	8	3	Yes	1
161	Male	29	Kolkata	Student	3	0	8.74	4	0	5-6 hours	Moderate	B.Ed	Yes	1	1	No	0
162	Male	29	Kolkata	Student	3	0	6.73	3	0	7-8 hours	Moderate	M.Tech	No	0	1	No	0
166	Female	25	Ahmedaba	Student	3	0	5.57	3	0	More than 8 hour	Unhealthy	MSc	Yes	10	5	No	1
172	Male	23	Thane	Student	1	0	8.59	4	0	7-8 hours	Healthy	BHM	No	11	3	No	0
173	Male	18	Bangalore	Student	4	0	7.1	3	0	More than 8 hour	Unhealthy	Class 12	Yes	11	5	Yes	1
176	Female	20	Mumbai	Student	5	0	8.58	5	0	7-8 hours	Moderate	Class 12	No	2	2	Yes	1
188	Male	31	Ahmedaba	Student	2	0	6.08	5	0	7-8 hours	Moderate	LLB	Yes	3	3	Yes	1

Image 1: Data Overview Table Snapshot

b. Data Collection Process:

The dataset was gathered through an online survey distributed to students across multiple programs and cities, capturing both academic and personal lifestyle data.

c. Data Limitations and Biases:

1. **Self-Reported Data:** Variables like sleep, diet, and depression status were self-reported, which may introduce bias due to inaccurate or socially desirable responses.
2. **Single-Year Coverage:** The dataset only reflects 2022, limiting trend or longitudinal analysis across multiple years.
3. **Missing Socioeconomic Context:** Important variables such as income, family background, or access to support services are absent, restricting deeper behavioral insights.

4. Limited Clinical Validation: Depression labels are based on survey responses, not professional diagnosis, meaning findings reflect perceived rather than confirmed depression.
5. Geographical and Program Imbalance: Representation across cities and degree types is uneven, which may skew certain comparisons or overemphasize some academic areas.

d. Suggestions

To strengthen future studies and institutional interventions:

- Collect multi-year data to track trends over time and measure intervention effectiveness.
- Incorporate socioeconomic and lifestyle factors for a more holistic understanding of student well-being.
- Collaborate with mental health professionals to validate depression metrics.
- Use balanced sampling across cities, genders, and degree programs to ensure fair representation.
- Integrate anonymous follow-up surveys to monitor post-intervention progress and reduce response bias.

DATA PREPARATION AND PREPROCESSING

Using Microsoft Excel, several cleaning and transformation steps were performed to prepare the dataset for analysis:

a. Data Cleaning

- Removed duplicate and blank entries (none found in this dataset).
- Standardized column names and applied consistent formatting for numerical variables.

b. Feature Engineering

To enhance interpretability:

1. Age Groups:

Formula: =IFS([@Age]<=19,"Teenager", [@Age]<=24,"Young Adult",
[@Age]<64,"Middle Aged")

2. State of Depression:

Formula: =IF([@Depression]=1, "Depression", "No Depression")

3. Rounded CGPA to two decimal places for uniformity.

id	Gender	Age	City	Profession	Academic Pressu	Work Pressure	CGPA	Study Satisfac	Job Satisfactio	Sleep Duration	Dietary Habits	Degree	Have you ever	Work/Study	Financial Stres	Family His	Depression
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26	Male	31	Srinagar	Student	3	0	7.03	5	0	Less than 5 hour	Healthy	BA	No	9	1	Yes	0
30	Female	28	Varanasi	Student	3	0	5.59	2	0	7-8 hours	Moderate	BCA	Yes	4	5	Yes	1
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62	Male	31	Nashik	Student	2	0	8.38	3	0	Less than 5 hour	Moderate	LLB	Yes	2	5	No	1
83	Male	24	Nagpur	Student	3	0	6.1	3	0	5-6 hours	Moderate	Class 12	Yes	11	1	Yes	1
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186	Male	31	Ahmedaba	Student	2	0	6.08	5	0	7-8 hours	Moderate	LLB	Yes	3	3	Yes	1

Image II: Dataset before preprocessing

id	Gender	Age	Age Group	City	Profess	Academic	Work Pre	CGPA	Study Satisf	Job Satisfac	Sleep Duration	Dietary Habits	Degree	Have you c	Work/	Financial s	Family	Depres	State of Dep
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8	Female	24	Young adult	Bangalore	Student	2	0	5.90	5	0	5-6 hours	Moderate	BSc	No	3	2	Yes	0	No Depression
26	Male	31	Middle aged	Srinagar	Student	3	0	7.03	5	0	Less than 5 hours	Healthy	BA	No	9	1	Yes	0	No Depression
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32	Female	25	Middle aged	Jaipur	Student	4	0	8.13	3	0	5-6 hours	Moderate	M.Tech	Yes	1	1	No	0	No Depression
33	Male	29	Middle aged	Pune	Student	2	0	5.70	3	0	Less than 5 hours	Healthy	PhD	No	4	1	No	0	No Depression
52	Male	30	Middle aged	Thane	Student	3	0	9.54	4	0	7-8 hours	Healthy	BSc	No	1	2	No	0	No Depression
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186	Male	31	Middle aged	Ahmedabi	Student	2	0	6.08	5	0	7-8 hours	Moderate	LLB	Yes	3	3	Yes	1	Depression
193	Male	25	Middle aged	Lucknow	Student	3	0	7.25	3	0	More than 8 hours	Unhealthy	M.Ed	Yes	10	5	No	1	Depression
208	Male	33	Middle aged	Indore	Student	5	0	5.74	2	0	Less than 5 hours	Moderate	M.Pharm	No	8	3	Yes	0	No Depression
214	Male	28	Middle aged	Kalyan	Student	3	0	9.85	3	0	7-8 hours	Unhealthy	M.Pharm	Yes	11	2	No	1	Depression
222	Male	18	Teenager	Surat	Student	4	0	6.70	5	0	Less than 5 hours	Moderate	Class 12	Yes	5	4	Yes	1	Depression
232	Male	18	Teenager	Visakhapa	Student	2	0	6.21	3	0	5-6 hours	Unhealthy	Class 12	Yes	4	2	No	1	Depression

Image III: Dataset after preprocessing

c. Industry Context:

Higher education—with focus on student wellness, academic performance, and institutional policy design.

d. Stakeholder Value:

The analysis helps universities identify high-risk student groups and key depression triggers, supporting data-driven interventions to improve both academic and mental health outcomes.

PRE-ANALYSIS

a. Key Trends Identified

The following dashboard visualizes these patterns observed below:

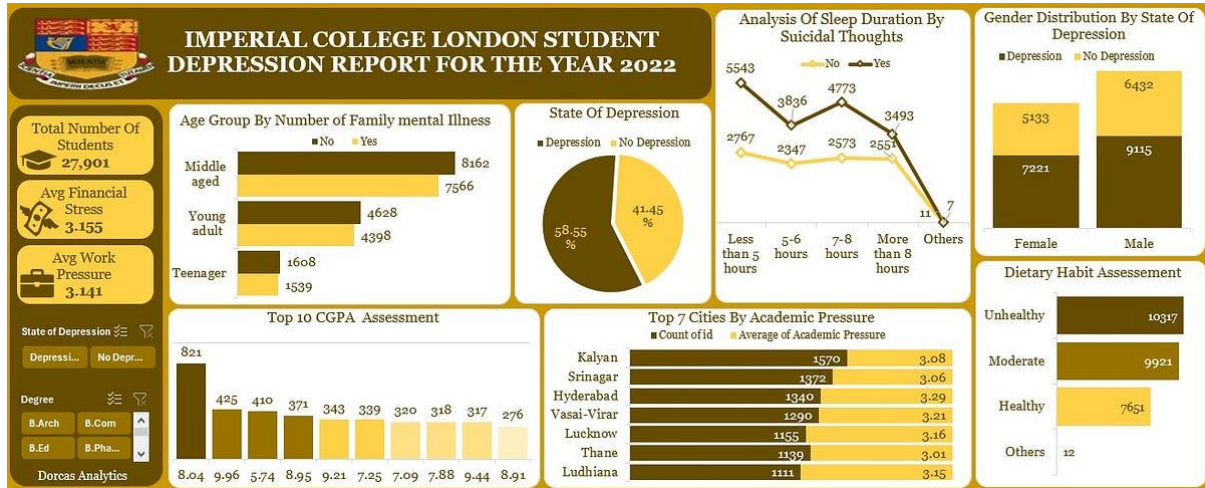


Image IV: Student Depression Report Dashboard

- Depression is widespread, affecting more than half of the student population (58.55%).
- Middle-aged and young-adult students report the highest depression rates, especially those balancing academic and family responsibilities.
- Sleep duration appears strongly associated with mental health—students sleeping fewer than 5 hours show higher depressive tendencies.
- Male students record slightly higher depression rates than females, challenging the common assumption that women report more emotional distress.
- Dietary habits seem to play a silent but important role—students with unhealthy or moderate diets show elevated depression levels.
- Academic pressure is notably intense in cities like Vasai-Virar, Kolkata, Ludhiana, Hyderabad, and Patna, coinciding with higher depression counts.

b. Potential Correlations

- Preliminary checks revealed potential relationships that warranted closer analysis:
- Academic Pressure ↔ Depression: Students facing higher academic stress consistently show poorer mental well-being.

- Sleep Duration ↔ Suicidal Thoughts: Insufficient sleep correlates with higher rates of suicidal ideation among depressed students.
- Family Mental Illness ↔ Age Group: Middle-aged students with a family history of mental illness show the greatest vulnerability.
- Gender ↔ Depression: Male students in high-pressure programs (especially B.Pharm and B.Com) are more likely to report depression.
- Diet ↔ Emotional Health: Healthier diets correspond to lower depression risk, while moderate or unhealthy diets coincide with higher depression levels.
- CGPA ↔ Depression: Surprisingly, high-performing students (CGPA between 8.0–9.9) are not immune; performance pressure may mask distress.

c. Initial Insights

From this early exploration, it became evident that academic performance alone doesn't define student wellness. Lifestyle factors such as sleep, diet, and family background interact with academic stress to shape emotional outcomes.

These insights suggested that effective interventions must go beyond grades—addressing personal well-being, peer influence, and workload balance to build resilient academic communities.

IN-ANALYSIS

a. Unconfirmed Insights and Recommendations

During the analytical stage, several patterns emerged that—while notable—require further validation:

- Age and Family Mental Illness: Middle-aged and young-adult students show high family mental illness rates. Universities could survey whether financial or family responsibilities contribute to increased emotional strain.
- State of Depression: With over half of students depressed, institutions should test the impact of structured counseling and regular wellness programs on academic performance.
- Gender Distribution: Male students appear disproportionately affected; further investigation could explore whether societal expectations or academic competition drive this.

- CGPA Patterns: High-CGPA students (8.04–9.96) still report depression, suggesting that internalized pressure may outweigh academic success. A longitudinal study could assess how stress evolves across semesters.

b. Recommendations

Based on in-depth analysis, several actionable steps are proposed:

- Mental Health Integration: Make counseling sessions compulsory across all departments to promote early intervention.
- Physical Activity and Lifestyle Programs: Encourage daily exercise, mindfulness activities, and nutrition awareness to relieve academic stress.
- Gender-Sensitive Support: Develop male-focused outreach programs addressing stigma around emotional expression.
- Monitoring Systems: Implement digital dashboards that help track academic workload, peer influence, and emotional triggers.
- Fair Institutional Practices: Ensure equality in policies across race, gender, and programs to reduce systemic stressors.
- Awareness and Education: Train lecturers and academic advisors to identify early signs of burnout or depression among students.

c. Analysis Techniques Used in Excel

The following Excel-based techniques were applied throughout the study:

- Data Cleaning: Removal of duplicates, standardization of column names, and formatting for uniform analysis.
- Feature Engineering: Creation of derived variables such as Age Group and Depression State using nested IFS and IF formulas.
- Pivot Tables: Used extensively to summarize categorical relationships (e.g., Gender vs. Depression, Age vs. Family Illness).
- Conditional Formatting: Highlighted depression clusters and academic pressure zones.
- Charts and Dashboards: Developed visual summaries including bar charts, pie charts, and dynamic filters to illustrate patterns and trends.

DATA VISUALIZATIONS & CHARTS

a. Key Visuals Developed in Excel:

- Gender Distribution by Depression: Displayed with stacked column chart.
- CGPA vs. total students: Displayed with a column chart.
- Age Group vs. Family Mental Illness: Displayed with Clustered bar chart.
- Academic Pressure by City and Degree: Displayed with 100% stacked bar chart.
- Sleep Duration vs. Suicidal Thoughts: Displayed with a line chart.

b. Dashboards:

Interactive Excel dashboards were designed to highlight depression trends across age, gender, diet, and academic programs, allowing stakeholders to identify and act on high-risk zones.

RECOMMENDATIONS AND OBSERVATIONS

a. Actionable Steps:

- Establish integrated mental health policies across all faculties.
- Develop student well-being dashboards for continuous monitoring.
- Encourage peer mentorship and emotional literacy initiatives.
- Partner with health professionals for regular awareness campaigns.

b. Optimizations:

- Adjust academic schedules to allow for rest and self-care.
- Offer anonymous counseling and well-being feedback forms each semester.

c. Unexpected Outcomes:

- Contrary to stereotypes, male and high-achieving students show significant vulnerability to depression.
- City-level academic intensity strongly correlates with mental distress.

CONCLUSION

This independent analysis paints a sobering picture: academic pressure, sleep deprivation, and inadequate nutrition are silently fueling depression across campuses. Depression is not confined to struggling students—even top-performing ones are at risk. Male and middle-aged students emerge as silent yet vulnerable groups.

To move forward, universities must redefine success—shifting focus from grades to holistic well-being. Through data-driven policymaking and targeted mental health initiatives, institutions can create campuses where students thrive mentally, emotionally, and academically.

REFERENCES AND APPENDICES

1. Shodolamu Opeyemi, 2024. Kaggle: Student Depression Dataset. Available at: <https://www.kaggle.com/datasets/hopesb/student-depression-dataset> (Accessed: 14th October 2025).
2. Helen Jahn, 2025. *Best Practices for Age Group Classification: How to Define Age Brackets for Health and Longevity*. Available at: <https://healthxwire.com/best-practices-for-age-group-classification/> (Accessed: 14th October 2025).